DSA03-Natural Language Processing

List of Lab Program

14. Create a program in python to check for agreement in sentences based on a context-free grammar's rules.

Code:

```
import nltk
from nltk import CFG
from nltk.parse import RecursiveDescentParser
grammar = CFG.fromstring("""
S -> NP_SG VP_SG | NP_PL VP_PL
NP_SG -> 'he' | 'she' | 'boy'
NP_PL -> 'they' | 'boys'
VP_SG -> 'plays' | 'runs'
VP_PL -> 'play' | 'run'
""")
parser = RecursiveDescentParser(grammar)
sentence = input("Enter a sentence: ").lower().split()
if list(parser.parse(sentence)):
    print("Sentence is grammatically correct")
else:
    print("Sentence is grammatically incorrect")
```

Output:

```
Enter a sentence: she runs
Sentence is grammatically correct
```